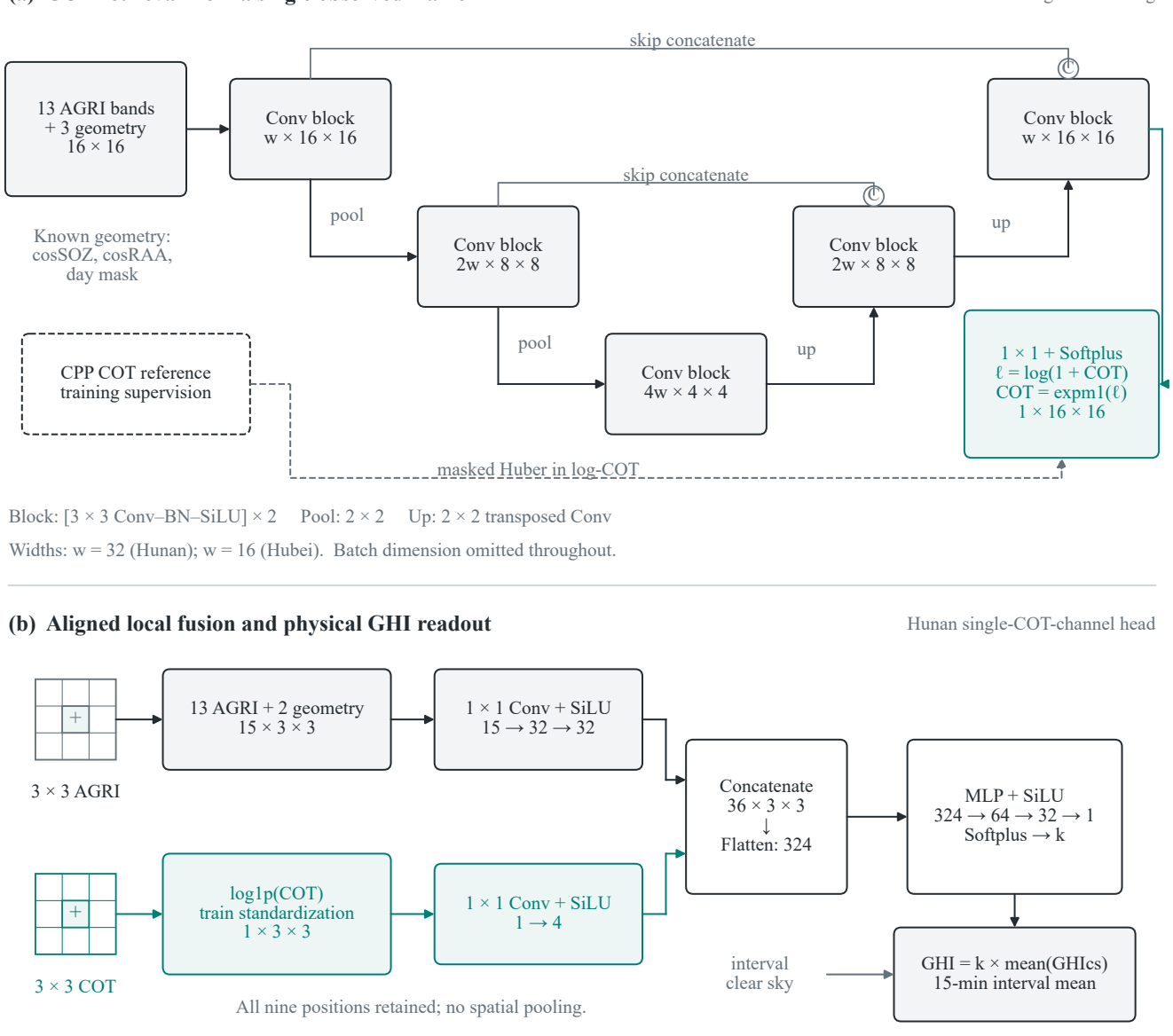

(a) COT retrieval from a single observed frame
R frozen during GHI training
skip concatenate
13 AGRI bands
+ 3 geometry
16 × 16
Conv block
w × 16 × 16
Conv block
w × 16 × 16
pool
skip concatenate
Known geometry:
cosSOZ, cosRAA,
day mask
Conv block
2w × 8 × 8
Conv block
2w × 8 × 8
up
pool
CPP COT reference
training supervision
Conv block
4w × 4 × 4
up
1 × 1 + Softplus
ℓ = log(1 + COT)
COT = expm1(ℓ)
1 × 16 × 16
masked Huber in log-COT
Block: [3 × 3 Conv–BN–SiLU] × 2 Pool: 2 × 2 Up: 2 × 2 transposed Conv
Widths: w = 32 (Hunan); w = 16 (Hubei). Batch dimension omitted throughout.
(b) Aligned local fusion and physical GHI readout
Hunan single-COT-channel head
13 AGRI + 2 geometry
15 × 3 × 3
1 × 1 Conv + SiLU
15 → 32 → 32
3 × 3 AGRI
Concatenate
36 × 3 × 3
↓
Flatten: 324
MLP + SiLU
324 → 64 → 32 → 1
Softplus → k
log1p(COT)
train standardization
1 × 3 × 3
1 × 1 Conv + SiLU
1 → 4
3 × 3 COT
interval
clear sky
GHI = k × mean(GHIcs)
15-min interval mean
All nine positions retained; no spatial pooling.
(c) Spatial order and implementation-specific controls
1 2 3
4 5 6
7 8 9
1 2 3
4 5 6
7 8 9
Same station pixels
Flatten: channel → row → column
Hubei four-arm interface
[zCOT, zPcloud] → Conv 2→4 (no bias)
COT-only arm: [zCOT, 0]